

Lab 10: Implementation of Diffie–Hellman Key Exchange

Course: Information Security & Cryptography (AI Department, SVNIT Surat)

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1. Introduction

The Diffie–Hellman (DH) key exchange protocol is a fundamental cryptographic technique that allows two parties to securely establish a shared secret key over an insecure communication channel. The security of this algorithm relies on the mathematical difficulty of solving the Discrete Logarithm Problem (DLP). Once a shared key is established, it can be used for symmetric encryption to ensure confidentiality of future communication between the two parties.

2. Algorithm Explanation

Let p be a large prime number and g a primitive root modulo p . The Diffie–Hellman algorithm proceeds as follows: Alice selects a private key a and computes her public value $A = g^a \bmod p$. Bob selects a private key b and computes his public value $B = g^b \bmod p$. Alice and Bob exchange their public values A and B over the insecure channel. Alice computes the shared key $K_A = B^a \bmod p$. Bob computes the shared key $K_B = A^b \bmod p$. Because $K_A = K_B = g^{ab} \bmod p$, both parties derive the same secret key without directly transmitting it.

3. Implementation Details

The implementation is written in C++ using 64-bit unsigned integers for arithmetic operations. Modular exponentiation is performed using the repeated squaring method, which efficiently computes $(\text{base}^{\text{exp}}) \bmod p$ in logarithmic time. Random numbers for Alice's and Bob's private keys are generated using the `std::mt19937_64` random number engine.

4. Sample Output (Screenshot Placeholder)

```
Public prime p = 4294967311
Generator g = 5
Alice private a = 2345214214
Alice public A = 3956268821
Bob private b = 1892643233
Bob public B = 1671163567
Shared key computed by Alice = 3191655838
Shared key computed by Bob = 3191655838
Keys match: YES
```

5. Security Explanation

An attacker observing the communication only knows the public parameters p , g , A , and B . To compute the shared key, the attacker must determine either Alice's private key a or Bob's private key b from their corresponding public values. This requires solving the **Discrete Logarithm Problem (DLP)**, which is computationally infeasible for large primes.

Therefore, even with access to all public data, the shared secret cannot be efficiently computed by an eavesdropper.

6. Conclusion

The Diffie–Hellman key exchange is one of the most important cryptographic protocols that established the foundation for modern public-key cryptography. Through this lab, we observed how two parties can securely generate a shared secret without directly sharing it over the network. Although the implementation uses smaller primes for demonstration, the same principle underlies real-world protocols such as TLS and VPN encryption. This experiment demonstrates the power of mathematical problems like discrete logarithms in ensuring data security.